

Project Name

InternMatch AI

AGILE PROJECT DOCUMENTATION

Project Title

**AI-Powered Internship Matching and Interview
Preparation Platform**

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Mentor

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Index

S. No	Section	Page No
01	Project Overview	01 – 02
02	Project Objective	03 – 04
03	Technology Stack	05 – 06
04	Project Requirement	07 – 09
05	System Architecture / Workflow	10 – 13
06	Product Backlog	14
07	Sprint Planning	15 – 17
08	Sprint Tasks and Status	18
09	Implementation / Key Features	19 – 30
10	Testing	31 – 33
11	Challenges and Solutions	34
12	Final Project Status	35 – 36

1. PROJECT OVERVIEW

1.1 Project Name

InternMatch AI

1.2 What is the Project?

InternMatch AI is a full-stack, AI-powered career assistance platform designed to help students and internship seekers throughout their internship and interview preparation journey.

The platform combines **resume intelligence, internship matching, skill-gap analysis, application management, AI-generated cover letters, AI assistance, interview preparation, mock interviews, voice interviews, and document-aware preparation chat** into a single application.

The system starts with the user's resume. The uploaded resume is processed using text extraction, Regex-based extraction, and Groq-powered Large Language Model processing. The extracted information is converted into a structured resume profile containing details such as skills, education, work experience, projects, certifications, internships, achievements, and other relevant information.

Based on the user's profile, the system helps identify suitable internships and provides information about matching skills and missing skills. Users can save internships, apply for them, generate personalized cover letters, and track their applications.

The platform also contains an AI Assistant and a dedicated Preparation Agent that help users prepare for interviews. The Preparation Agent provides learning recommendations, interview questions, answer evaluation, mock interviews, voice interviews, progress tracking, and final preparation results.

The project uses a separate **React frontend** and **FastAPI backend**, with **PostgreSQL** used for persistent storage and **SQLAlchemy** used for database access. Groq is integrated to provide LLM-powered AI functionality.

1.3 Problem Statement

Students and fresh graduates often face difficulties in finding internships that match their skills and preparing effectively for interviews.

Traditional internship platforms mainly require users to manually search through large numbers of internship listings. Students may not know whether their existing skills are sufficient for a particular internship or which skills they need to improve.

Similarly, interview preparation platforms often provide generic questions without considering the candidate's actual resume, skills, projects, education, and experience.

There is therefore a need for an integrated career platform that can:

- Understand a user's resume.
- Identify the user's skills and qualifications.
- Find relevant internship opportunities.
- Identify missing skills for selected internships.
- Assist with internship applications.
- Generate personalized cover letters.
- Provide AI-based career assistance.
- Provide personalized interview preparation.
- Support mock and voice interviews.
- Allow users to ask questions about uploaded preparation documents.

InternMatch AI addresses these problems by combining resume analysis, internship matching, application assistance, and AI-powered interview preparation in one platform.

1.4 Target Users

The primary target users are:

- Students searching for internships.
- Fresh graduates.
- Internship seekers.
- Candidates preparing for technical interviews.
- Candidates preparing for HR interviews.
- Students who need help identifying skill gaps.
- Users who need personalized interview preparation.
- Users who want AI assistance during their career preparation.

2. PROJECT OBJECTIVE

The main objective of **AI Career Companion Agent for Internship Matching and Interview Preparation** is to develop an intelligent and personalized platform that helps students and internship seekers throughout their career journey. The system integrates **resume analysis, internship matching, skill-gap analysis, application management, AI assistance, and interview preparation** into a single platform.

Objectives

1. Resume Intelligence

To automatically analyze uploaded resumes and convert unstructured resume content into a structured candidate profile. The system supports PDF, DOC, and DOCX files and uses **text extraction, Regex-based parsing, and Groq LLM processing** to identify important information such as skills, education, experience, projects, certifications, internships, achievements, and other candidate details.

2. Intelligent Internship Matching

To help users find internship opportunities that are relevant to their qualifications and skills. The system uses the candidate's resume profile and internship requirements to provide more personalized internship recommendations.

3. Skill-Gap Analysis

To identify the difference between the skills a candidate currently possesses and the skills required for a selected internship. This helps users understand their weaknesses and focus on the skills they need to improve.

4. Internship Application Support

To provide a centralized platform for managing internship applications. Users can view internship details, save internships, apply for opportunities, view application information, and withdraw applications when required.

5. AI-Powered Cover Letter Generation

To generate personalized cover letters based on the user's resume information and selected internship details. The system aims to create relevant application content using the candidate's actual skills, projects, education, and experience.

6. AI Career Assistance

To provide an AI Assistant that helps users with internship, skill, application, and career-related questions. The assistant can maintain conversations and provide responses using available candidate and platform context.

7. Personalized Interview Preparation

To provide an interactive **Preparation Agent** for interview preparation. The system supports interview question generation, answer evaluation, mock interviews, voice-based interviews, learning recommendations, progress tracking, and preparation results.

8. Document-Based Preparation

To allow users to upload preparation documents and interact with them through the **Preparation Chat**. Users can ask questions related to their uploaded study material and continue their preparation through conversation history.

9. Personalized Career Experience

To use the candidate's own resume and profile information across different modules, creating a connected career experience from **resume analysis to internship matching, application support, and interview preparation**.

3. TECHNOLOGY STACK

InternMatch AI is developed using a full-stack architecture consisting of modern frontend, backend, database, and AI technologies.

Category	Technology	Purpose
Frontend	React.js	Build the interactive user interface and application.
Build Tool	Vite	Provides fast frontend development and build configuration.
Backend	Python + FastAPI	Develops REST API's and handles application logic.
Database	PostgreSQL	Stores users, resumes, internships, applications, conversations, and other application data.
ORM	SQLAlchemy	Connects the FastAPI backend with PostgreSQL and manages database operations.
AI / LLM	Groq	Provides LLM – based resume analysis, AI assistance, cover – letter generation, and interview preparation.
Resume Processing	PyMuPDF	Extracts text from PDF resumes and documents.
Document Processing	Python-docx	Processes DOCX documents and generates DOCX files.
Authentication	JWT + bcrypt	Provides secure authentication and password hashing.
API Communication	Axios	Connects the React frontend with FastAPI backend API's.
Icons	Lucide React	Provides icons for the application's user interface.
Version Control	Git + GitHub	Maintain source code and project version history.

Technology Overview:

The **React.js frontend** provides the user interface for registration, login, resume management, internships, applications, AI assistance, and interview preparation.

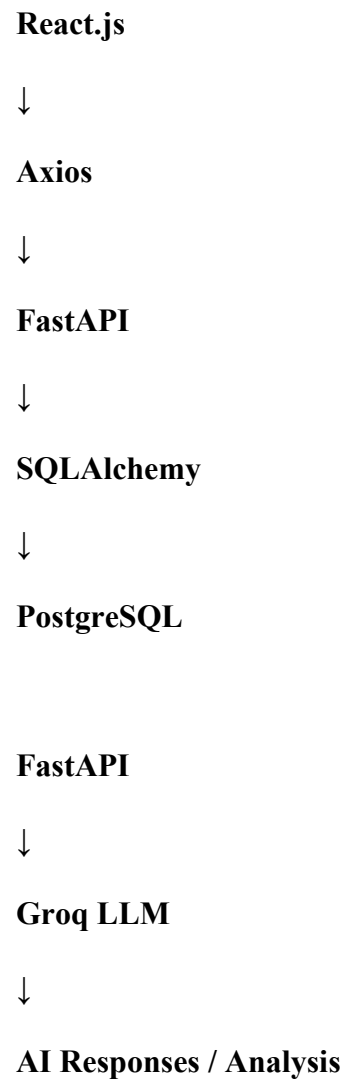
The **FastAPI backend** manages authentication, resume processing, internships, applications, AI services, preparation features, and communication with the database.

PostgreSQL is used for persistent storage, while **SQLAlchemy** provides the database interaction layer.

Groq is integrated into the application to provide AI capabilities such as resume analysis, conversational assistance, cover-letter generation, and interview preparation.

For document processing, **PyMuPDF** is used for PDF text extraction and **python-docx** is used for DOCX processing.

Overall technology flow:



4. PROJECT REQUIREMENT

The project requirements define the major functions and qualities required for **InternMatch AI** to provide a complete internship-matching and interview-preparation experience. The requirements are divided into **Functional Requirements** and **Non-Functional Requirements**, following the structure of the provided documentation format.

Functional Requirements

The system should provide the following major functionalities:

1. User Authentication

- Users should be able to register and log in securely.
- The system should authenticate users using JWT-based authentication.
- Users should be able to manage their account and password.

2. Resume Management

- Users should be able to upload their resumes in supported formats such as PDF, DOC, and DOCX.
- The system should extract and analyze resume content.
- Resume information should be converted into a structured candidate profile.

3. Internship Management

- Users should be able to view available internships.
- Users should be able to view detailed internship information.
- The system should provide relevant internship opportunities based on the user's profile.

4. Skill Gap Analysis

- The system should compare candidate skills with internship requirements.
- Users should be able to identify missing or required skills.
- The system should help users understand areas that require improvement.

5. Internship Applications

- Users should be able to apply for internships.
- Users should be able to view and manage their applications.
- Users should be able to withdraw applications when required.

6. Saved Internships

- Users should be able to save internships for future reference.
- Users should be able to manage their saved internship list.

7. AI Cover Letter Generation

- The system should generate personalized cover letters based on the user's resume and selected internship.
- Users should be able to download generated cover letters in supported formats.

8. AI Assistant

- Users should be able to ask career and internship-related questions.
- The AI Assistant should provide responses using available user and platform context.
- Users should be able to maintain and manage their conversations.

9. Interview Preparation

- Users should be able to start interview preparation sessions.
- The system should generate relevant interview questions.
- The system should evaluate user answers.
- Users should be able to view preparation results and progress.

10. Mock and Voice Interviews

- Users should be able to participate in mock interviews.
- The system should generate mock interview questions and evaluate answers.
- Users should be able to practice interviews using voice-based interaction.

11. Preparation Chat

- Users should be able to create and manage preparation conversations.
- Users should be able to upload preparation documents.
- The AI should answer questions based on uploaded preparation material.
- Users should be able to view and delete conversation history.

12. Notifications

- The system should provide relevant notifications related to user activities and applications.

Non-Functional Requirements

1. Security

The system should protect user accounts and personal information through secure authentication, password hashing, and authenticated API access.

2. Performance

The application should provide reasonable response times for user interactions, database operations, document processing, and AI requests.

3. Usability

The interface should be simple, clear, and easy to navigate for students and internship seekers.

4. Reliability

The system should handle invalid inputs, authentication errors, document-processing errors, and API failures appropriately.

5. Scalability

The architecture should support the addition of more users, internships, AI capabilities, and preparation resources in the future.

6. Maintainability

The frontend and backend should be organized into separate modules and services so that individual features can be developed, tested, and maintained easily.

7. Data Management

User profiles, resumes, internships, applications, conversations, and preparation-related information should be stored and managed through the PostgreSQL database.

8. AI Response Quality

AI-generated results should be relevant to the user's available profile and should avoid generating unsupported candidate information.

5. SYSTEM ARCHITECTURE / WORKFLOW

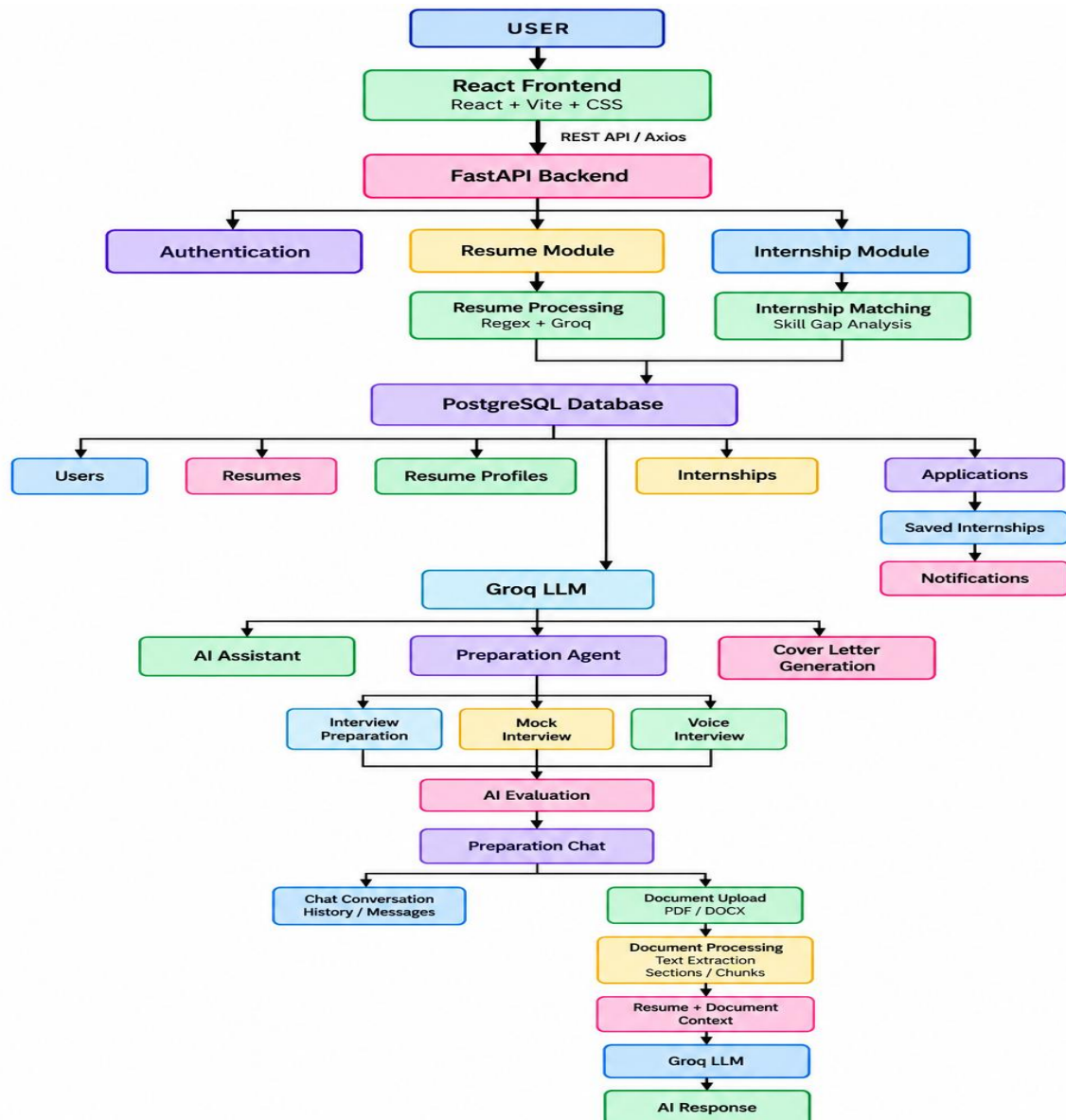
The **InternMatch AI** system follows a full-stack architecture in which the React frontend communicates with the FastAPI backend through APIs. The backend manages authentication, resume processing, internships, applications, AI services, and interview preparation, while PostgreSQL stores the application's data.

The major technologies and components work together as follows:

React Frontend → **FastAPI Backend** → **Application Services** → **PostgreSQL / Groq** → **Response** → **React Frontend**.

5.1 Architecture Diagram

Use the following diagram in your documentation:

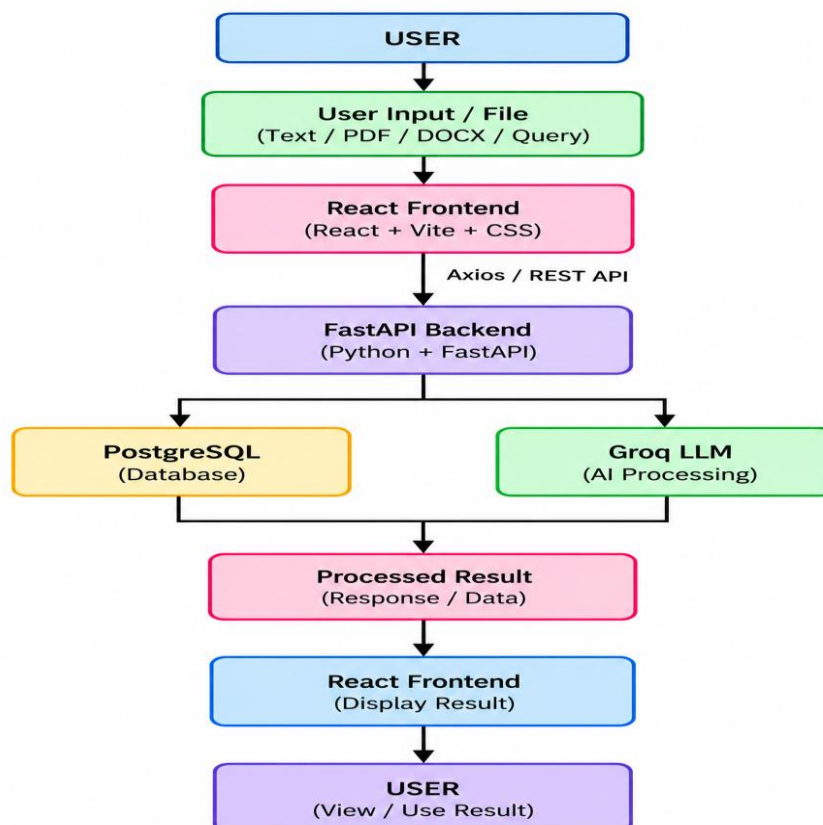


5.2 Main Components

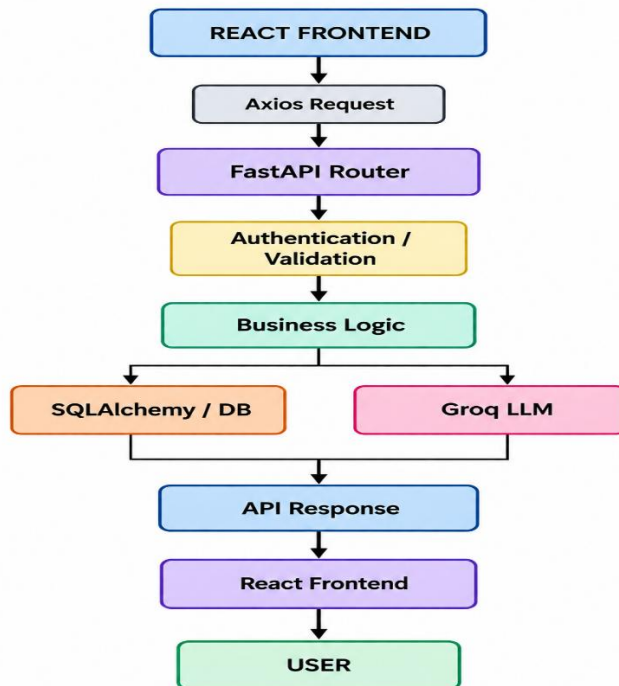
The major components of the system are:

1. **React Frontend** – Provides the user interface for authentication, resume management, internships, applications, AI Assistant, Preparation Agent, and Preparation Chat.
2. **FastAPI Backend** – Handles REST APIs, authentication, business logic, resume processing, internship operations, applications, AI services, and preparation features.
3. **PostgreSQL Database** – Stores users, resumes, resume profiles, internships, saved internships, applications, notifications, AI conversations, preparation conversations, documents, sections, chunks, and messages.
4. **Resume Processing Module** – Extracts text from uploaded PDF/DOCX resumes and uses Regex and Groq for structured resume analysis.
5. **Internship Module** – Provides internship discovery, resume-based matching, saved internships, and skill gap analysis.
6. **AI Services** – Uses Groq LLM for resume analysis, AI Assistant responses, cover-letter generation, interview preparation, and answer evaluation.
7. **Preparation Agent** – Supports interview preparation, mock interviews, voice interviews, question generation, and AI evaluation.
8. **Preparation Chat** – Provides conversational preparation using resume information and uploaded PDF/DOCX documents.

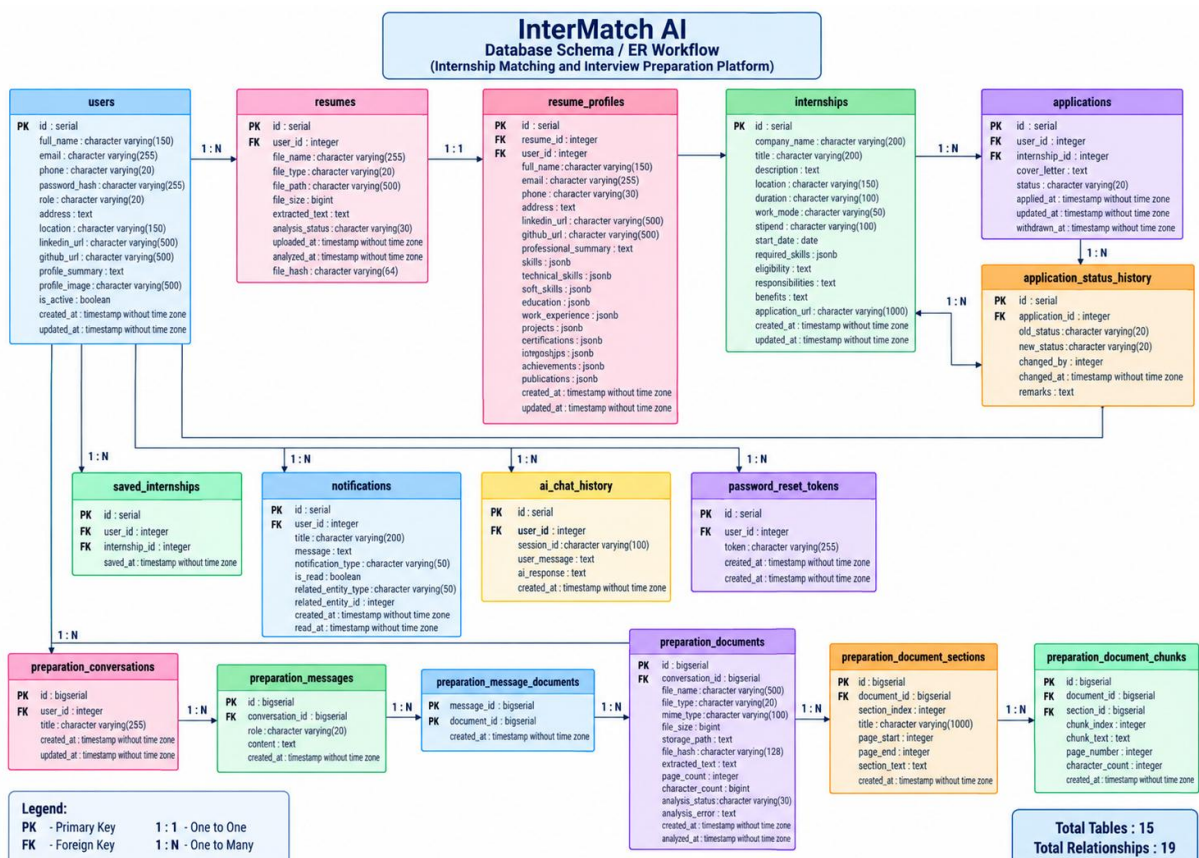
5.3 Data Flow Diagram



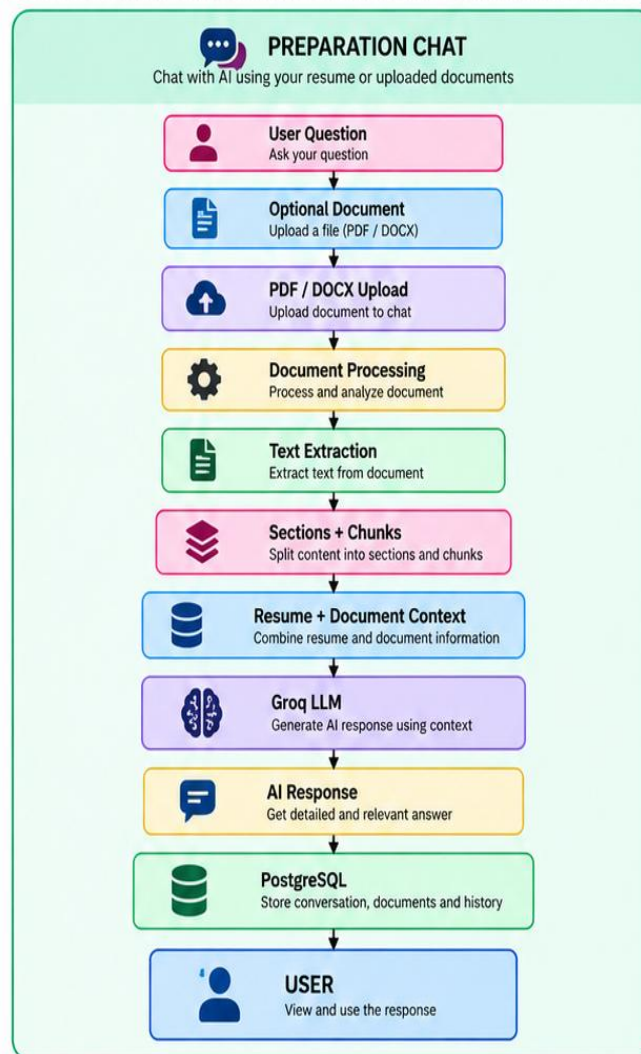
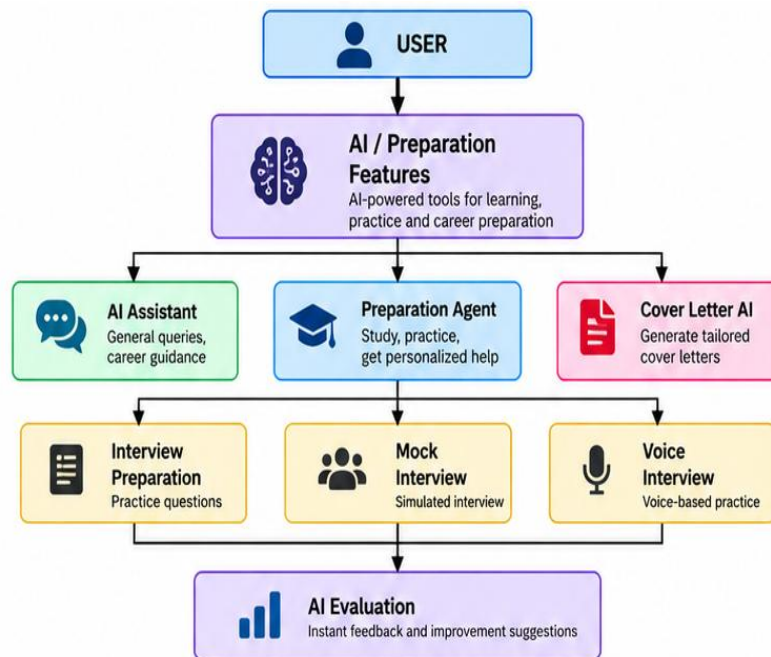
5.4 API Flow Diagram



5.5 Database Schema / Flow Diagram



5.6 AI / RAG / Agent Workflow Diagram



6. PRODUCT BACKLOG

Backlog Priority

- **High:** Core functionality required for the main internship matching and interview preparation workflow.
- **Medium:** Supporting features that improve usability and the overall user experience.

The Product Backlog contains the major features and development tasks required to build the **InternMatch AI** platform. The backlog is prioritized according to the importance of each feature to the overall system.

ID	Feature / Task	Priority	Status
PB - 01	User Registration and Login	High	Completed
PB - 02	JWT Authentication and Security	High	Completed
PB - 03	Password Recovery and Account Management	High	Completed
PB - 04	Resume Upload and Validation	High	Completed
PB - 05	Resume Text Extraction	High	Completed
PB - 06	Regex – Based Resume Parsing	High	Completed
PB - 07	Groq LLM Resume Analysis	High	Completed
PB - 08	Structured Resume Profile	High	Completed
PB - 09	Internship Discovery	High	Completed
PB - 10	Resume – Based Internship Matching	High	Completed
PB - 11	Skill Gap Analysis	High	Completed
PB - 12	Save Internships	Medium	Completed
PB - 13	Internship Application Management	High	Completed
PB - 14	AI Cover Letter Generation	High	Completed
PB - 15	Cover Letter PDF / DOCX Download	Medium	Completed
PB - 16	User Notifications	Medium	Completed
PB - 17	AI Assistant and Chat History	High	Completed
PB - 18	Preparation Agent	High	Completed
PB - 19	Mock Interview	High	Completed
PB - 20	Voice Interview	High	Completed
PB - 21	AI Answer Evaluation	High	Completed
PB - 22	Preparation Chat	High	Completed
PB - 23	Preparation Document Upload	High	Completed
PB - 24	Document Sections and Chunk Processing	High	Completed
PB - 25	Preparation Conversation History	Medium	Completed
PB - 26	Response Frontend and Reusable UI	Medium	Completed

7. SPRINT PLANNING

Sprint 1 — Goal: Project foundation and authentication [1st week]

- Set up React/Vite frontend and FastAPI backend structure.
- Configure PostgreSQL and SQLAlchemy connectivity.
- Implement user registration and login.
- Implement JWT authentication and protected routes.
- Implement password recovery and account management.
- Create core database models and persistence.

Sprint 2 — Goal: Resume intelligence [2nd week]

- Implement PDF/DOCX resume upload and validation.
- Implement resume text extraction using PyMuPDF and python-docx.
- Implement Regex-based resume parsing.
- Integrate Groq LLM for structured resume analysis.
- Create structured resume profiles.
- Store education, skills, experience, projects, certifications, and other resume information.

Sprint 3 — Goal: Internship discovery and matching [3rd week]

- Implement internship catalog and data handling.
- Implement internship search and filtering.
- Implement resume-based internship matching.
- Implement skill-based matching and internship match details.
- Implement skill gap analysis.
- Implement saved internship functionality.

Sprint 4 — Goal: Internship applications and career support [4th week]

- Implement internship application functionality.

- Implement application withdrawal and application status management.
- Implement applied internship management.
- Implement user notifications.
- Implement AI-generated personalized cover letters.
- Implement PDF/DOCX cover-letter downloads.

Sprint 5 — Goal: AI Assistant [5th week]

- Implement the AI Assistant interface.
- Implement authenticated user context for AI responses.
- Integrate Groq LLM for AI-generated responses.
- Implement persistent chat sessions and conversation history.
- Implement contextual career and internship-related assistance.
- Integrate the AI Assistant with the React frontend and FastAPI backend.

Sprint 6 — Goal: Complete Preparation Agent [6th week]

- Implement the separate Preparation Agent module.
- Implement personalized interview-preparation guidance.
- Implement interview question generation.
- Implement learning and preparation recommendations.
- Implement mock interview functionality.
- Implement voice interview functionality.
- Implement AI-based answer evaluation.
- Store interview sessions, questions, answers, and results.

Sprint 7 — Goal: Preparation Chat and document processing [7th week]

- Implement Preparation Chat with a ChatGPT-like interface.
- Implement preparation conversation history.
- Implement new and delete conversation functionality.
- Implement PDF/DOCX document upload.

- Implement document text extraction and processing.
- Implement document sections and chunks.
- Implement message-document relationships.
- Enable responses using resume information and uploaded document content.

Sprint 8 — Goal: System integration, UI refinement and final validation [8th week]

- Integrate all frontend routes and backend APIs.
- Refine reusable layouts, sidebars, forms, and responsive UI.
- Validate authentication and protected routes.
- Validate resume analysis and internship matching workflows.
- Validate application, cover-letter, AI Assistant, and Preparation Agent workflows.
- Perform final functional testing and bug fixing.
- Prepare final project documentation and demo materials.

8. SPRINT TASKS AND STATUS

Sprint	Task	Status
Sprint 1	Project setup, repository, backend and frontend	Done
Sprint 1	PostgreSQL, SQLAlchemy configuration and database persistence	Done
Sprint 1	User registration, Login, JWT authentication and protected routes	Done
Sprint 1	Password recovery and account management	Done
Sprint 2	PDF / DOCX resume upload, validation and text extraction	Done
Sprint 2	Regex – based and Groq LLM resume parsing	Done
Sprint 2	Structured resume profile creation and storage	Done
Sprint 2	Resume education, skills, experience, projects and certification extraction	Done
Sprint 3	Internship catalog, search and filtering	Done
Sprint 3	Resume – based Internship matching and match details	Done
Sprint 3	Skill gap analysis	Done
Sprint 3	Saved internships functionality	Done
Sprint 4	Internship application and withdrawn functionality	Done
Sprint 4	Application status management and notifications	Done
Sprint 4	AI – generated personalized cover letters	Done
Sprint 4	Cover – letter PDF / DOCX download	Done
Sprint 5	AI Assistant with Groq LLM integration	Done
Sprint 5	AI Assistant contextual career and internship assistance	Done
Sprint 5	AI Assistant chat sessions and conversation history	Done
Sprint 6	Preparation Agent implementation	Done
Sprint 6	Interview question generation and preparation guidance	Done
Sprint 6	Mock interview functionality	Done
Sprint 6	Voice interview and AI answer evaluation	Done
Sprint 6	Interview session and result management	Done
Sprint 7	Preparation chat and conversation management	Done
Sprint 7	Preparation document upload and processing	Done
Sprint 7	Document sections and chunk processing	Done
Sprint 7	Preparation message – document relationships and contextual response	Done
Sprint 8	Frontend and backend integration	Done
Sprint 8	UI refinement, reusable components and responsive layout	Done
Sprint 8	Final workflow validation, bug fixing and testing	Done
Sprint 8	Final Documentation and demo preparation	Done

9. IMPLEMENTATION / KEY FEATURES

9.1 User Registration and Login

Users can create an account and log in securely. JWT-based authentication is used to protect user-specific application routes.

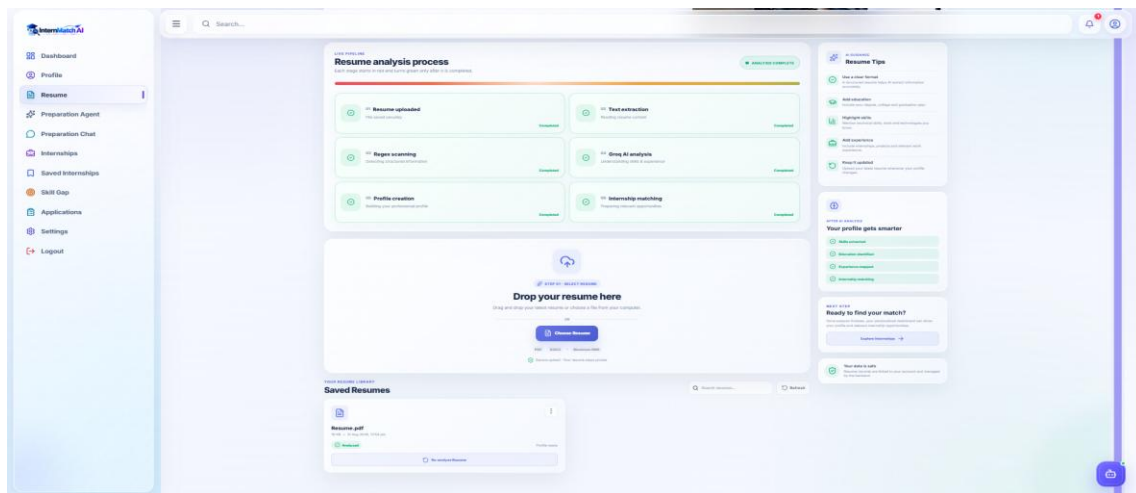
The image displays two side-by-side screenshots of the InternMatch AI user interface. The left screenshot shows the login page, featuring the InternMatch AI logo, a 'LOGIN TO YOUR ACCOUNT' button, a welcome message, and input fields for 'Email Address' and 'Password'. A 'Login' button is at the bottom. The right screenshot shows the registration page, featuring the InternMatch AI logo, a 'CREATE YOUR ACCOUNT' button, a message to 'Start your journey to find the right internship.', and input fields for 'Full Name', 'Email Address', 'Phone Number', 'Password', and 'Confirm Password'. A 'Create Account' button is at the bottom.

9.2 Resume upload and AI Analysis

Users upload their resume in PDF or DOCX format. The system extracts the resume text and processes it using Regex-based parsing and Groq LLM analysis to create a structured resume profile.

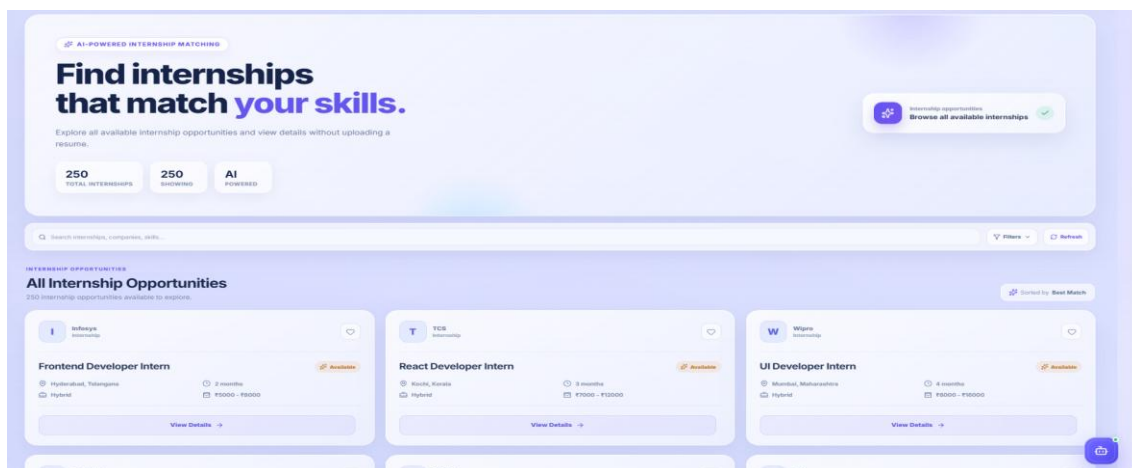
The image shows a screenshot of the InternMatch AI dashboard. The dashboard has a sidebar with navigation links: Dashboard, Profile, Resume, Preparation Agent, Preparation Chart, Internships, Saved Internships, Skill Gap, Applications, Settings, and Logout. The main content area features a 'Build a resume that gets noticed.' section with a sub-header 'AI Career Assistant' and a description. Below this is a 'Resume analysis process' section with a flowchart showing steps: Resume uploaded, Text extraction, Regex scanning, Groq AI analysis, Profile creation, and Internship matching. At the bottom, there is a 'Drop your resume here' section with a 'Choose Resume' button.

After uploading resume, resume page looks like this



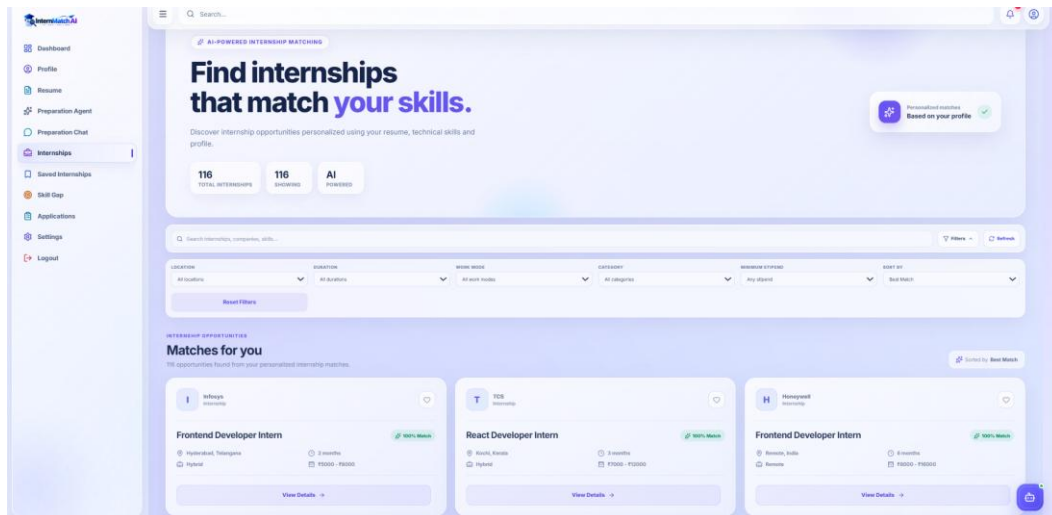
9.3 Internship Discovery and Search

Users can browse available internships and use search and filtering options to find relevant opportunities.

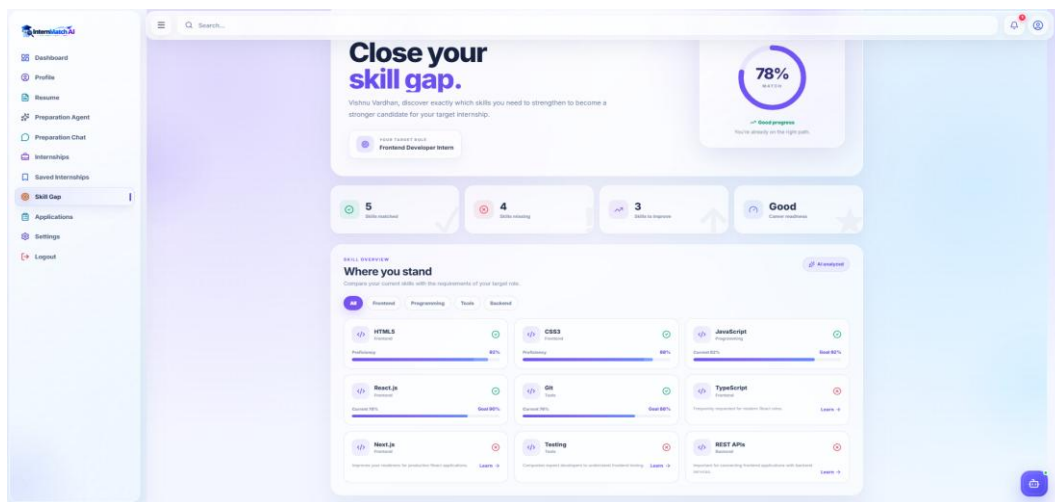


9.4 Internship Matching and Skill Gap Analysis

The system compares the user's resume profile with internship requirements and provides relevant matching information and skill-gap details.

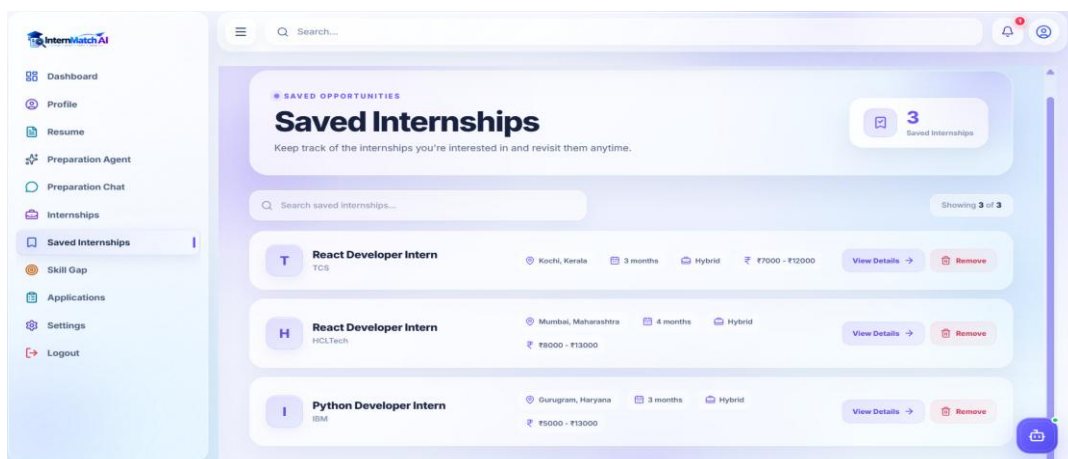


Skill Gap page



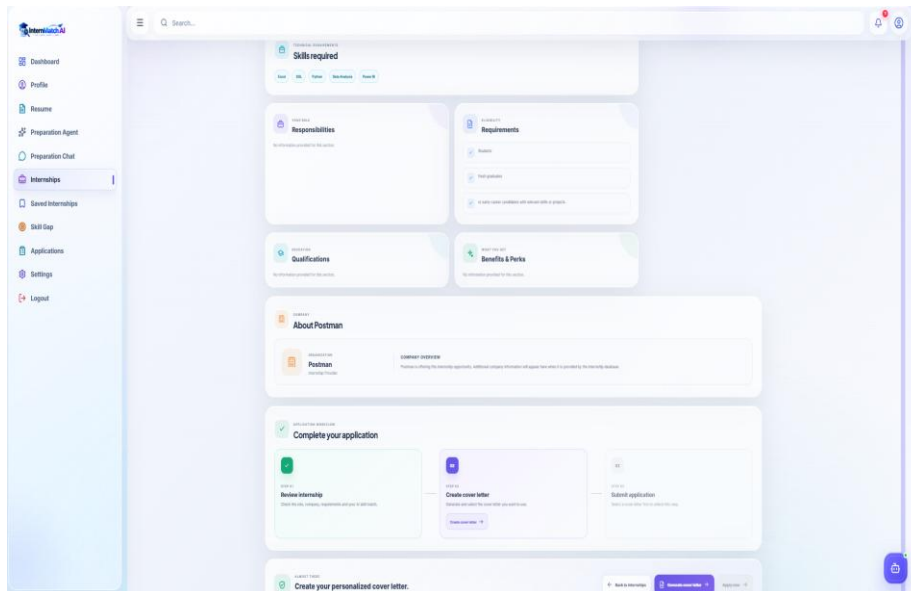
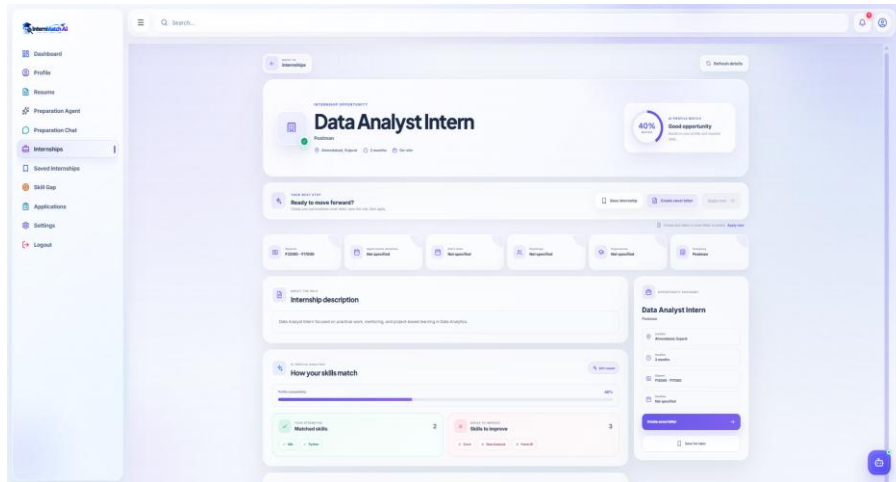
9.5 Saved Internships

Users can save internships for later reference and access their saved opportunities from the application.

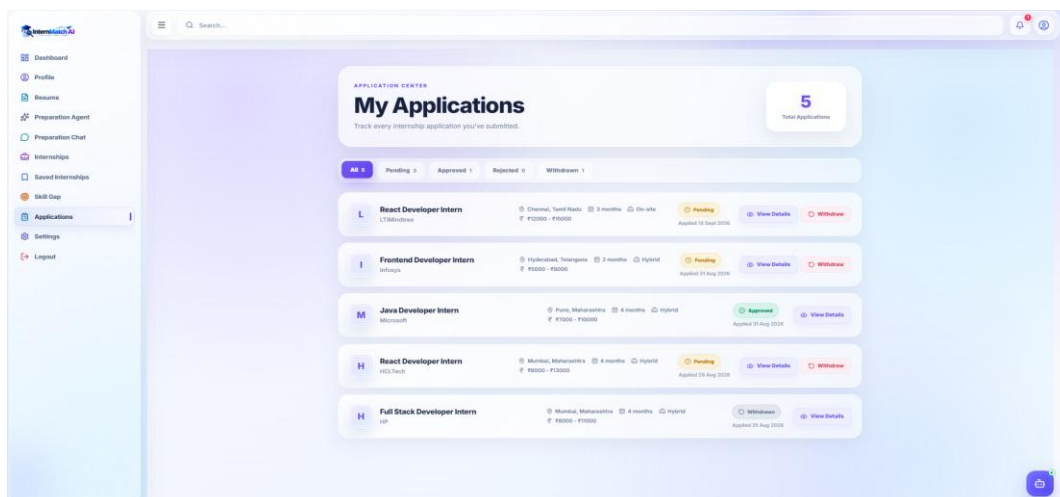


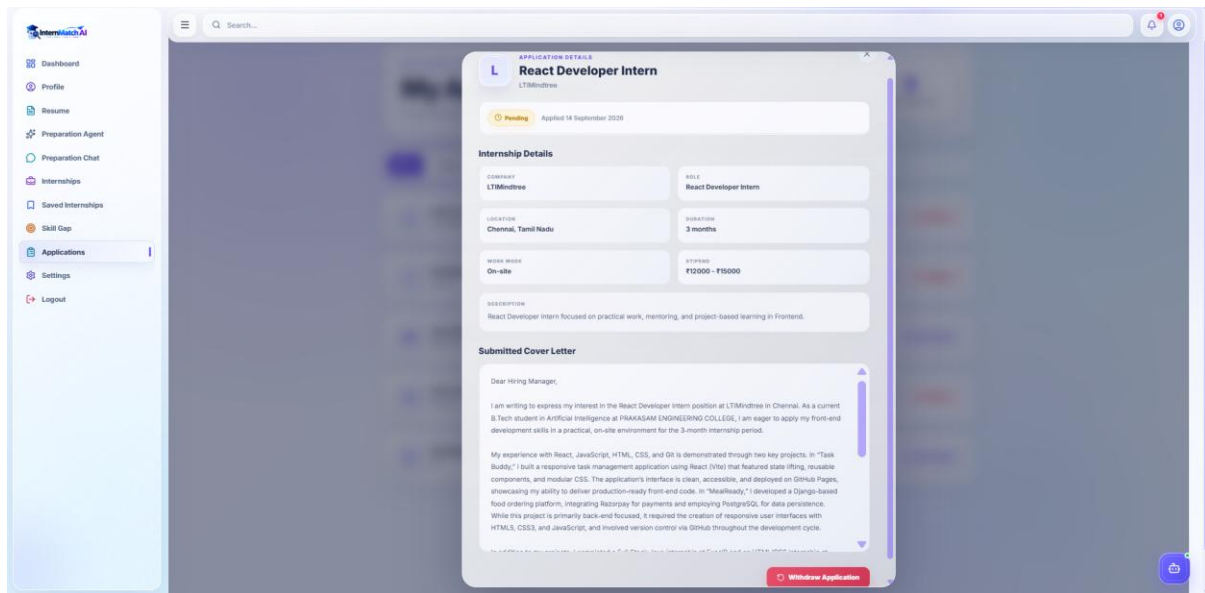
9.6 Internship Applications

Users can apply for internships, view their applications, and manage application-related actions such as withdrawal where supported.



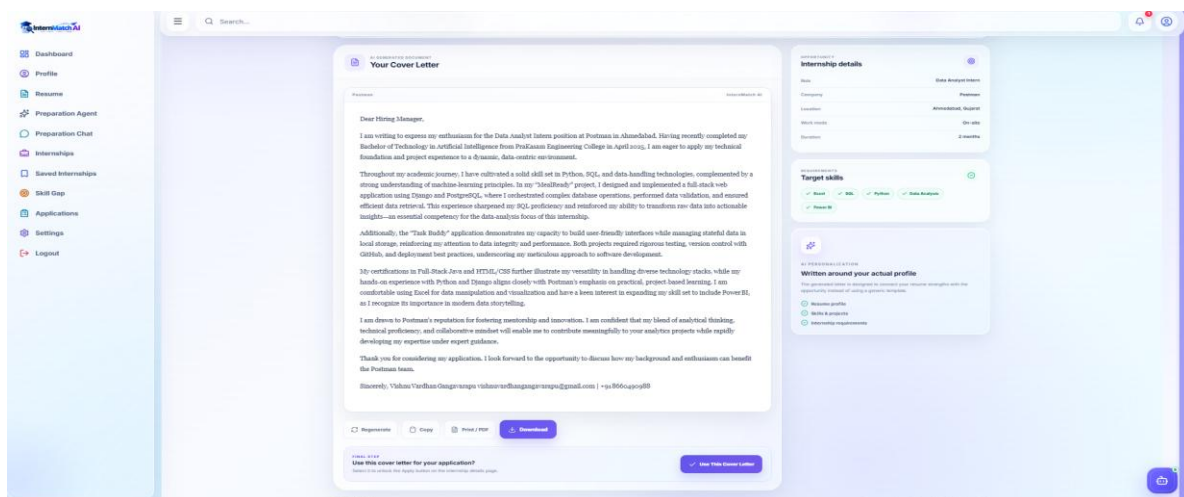
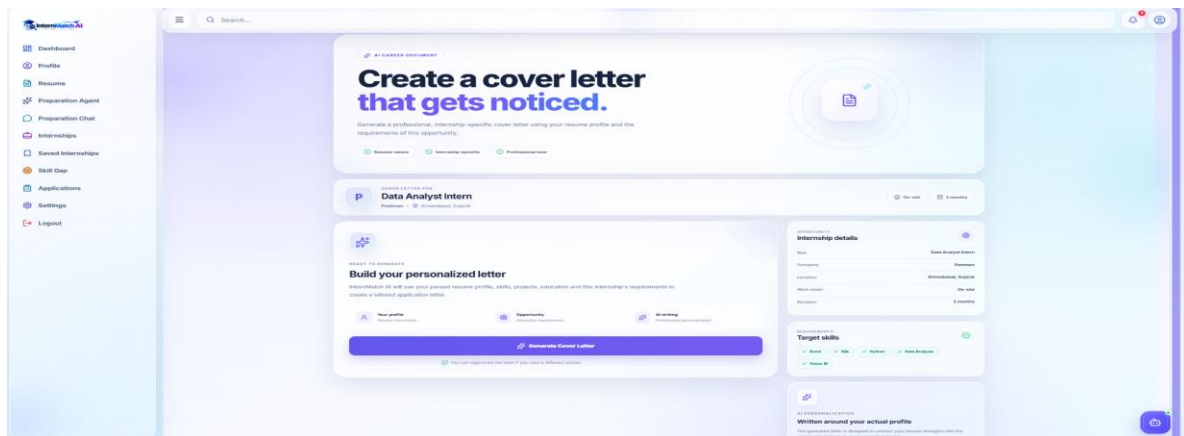
Applications Page





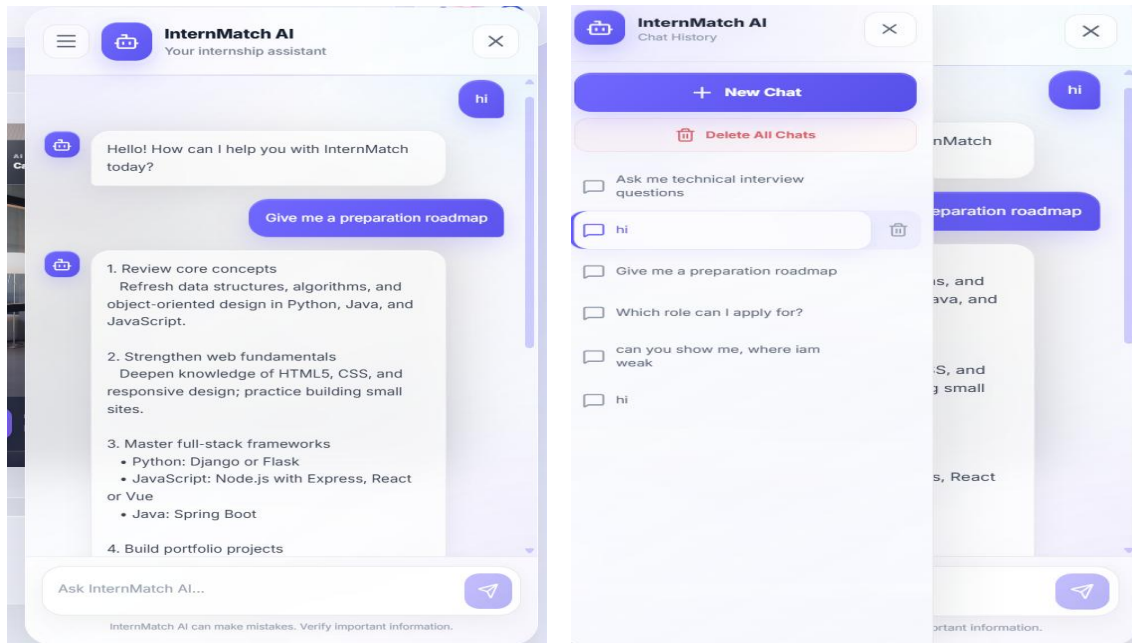
9.7 AI Cover Letter Generation

The system generates a personalized cover letter using the user's resume information and internship-related context.



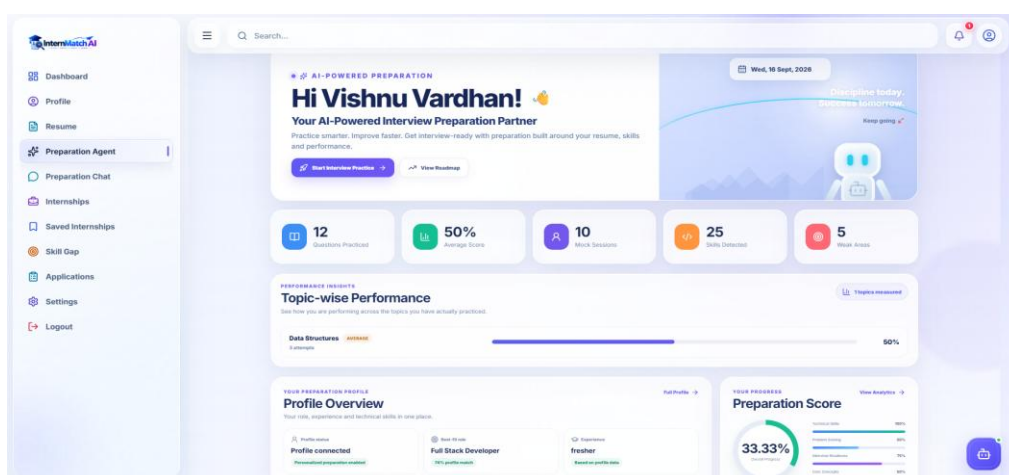
9.8 AI Assistant

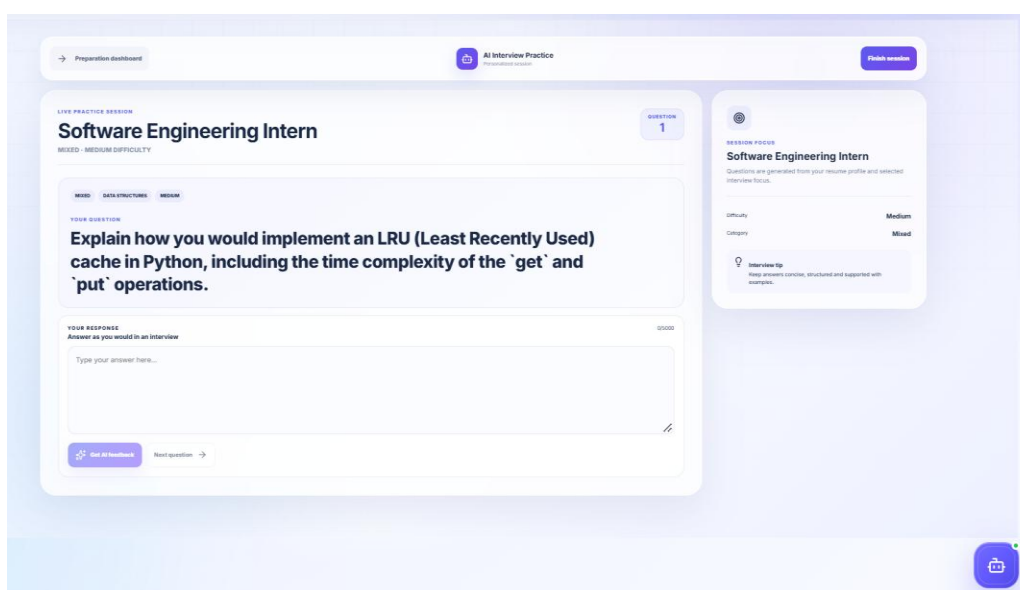
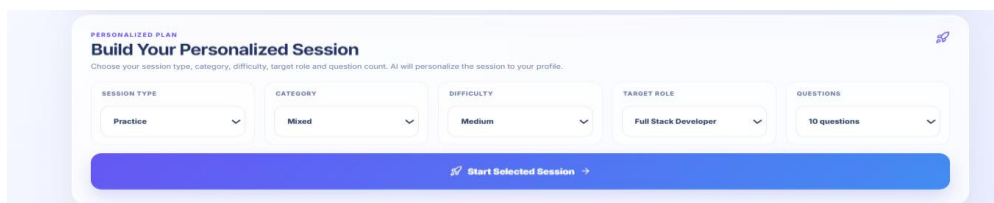
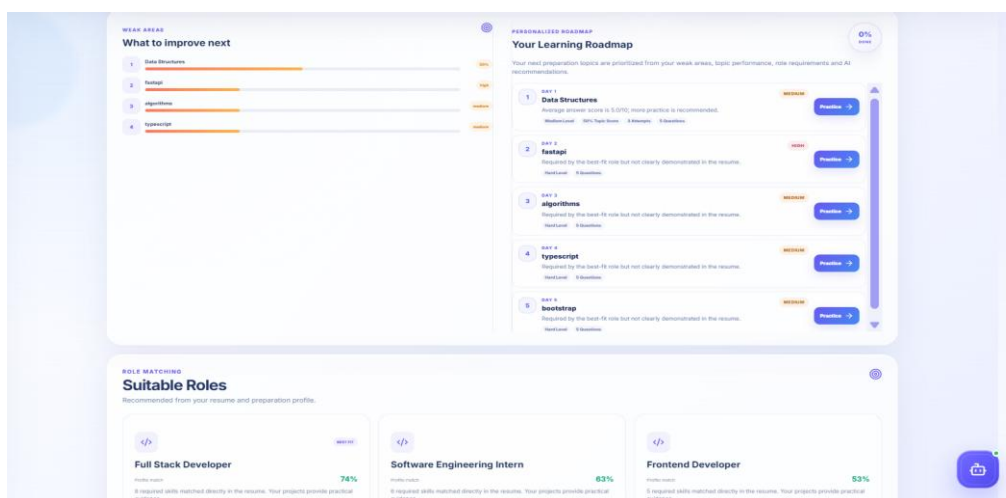
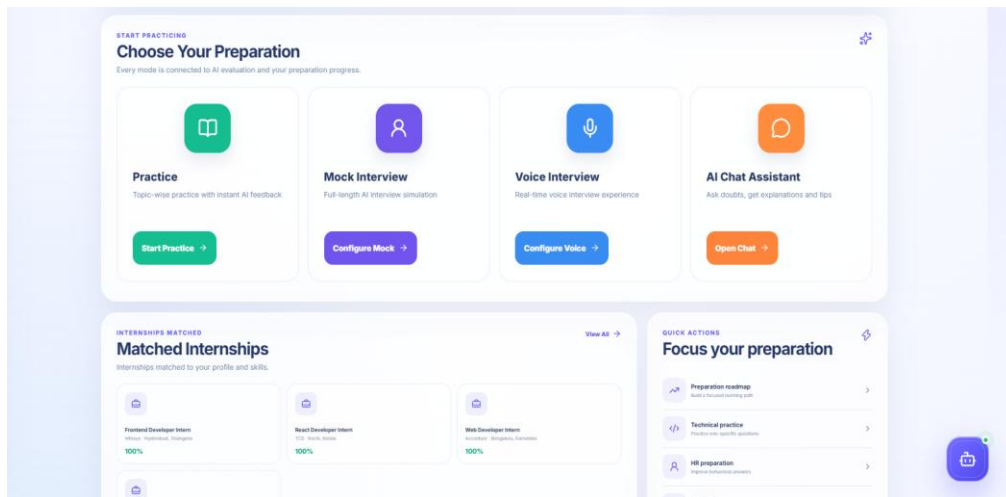
The AI Assistant uses authenticated user and platform context to answer career and internship-related questions. Conversations are maintained through chat history.



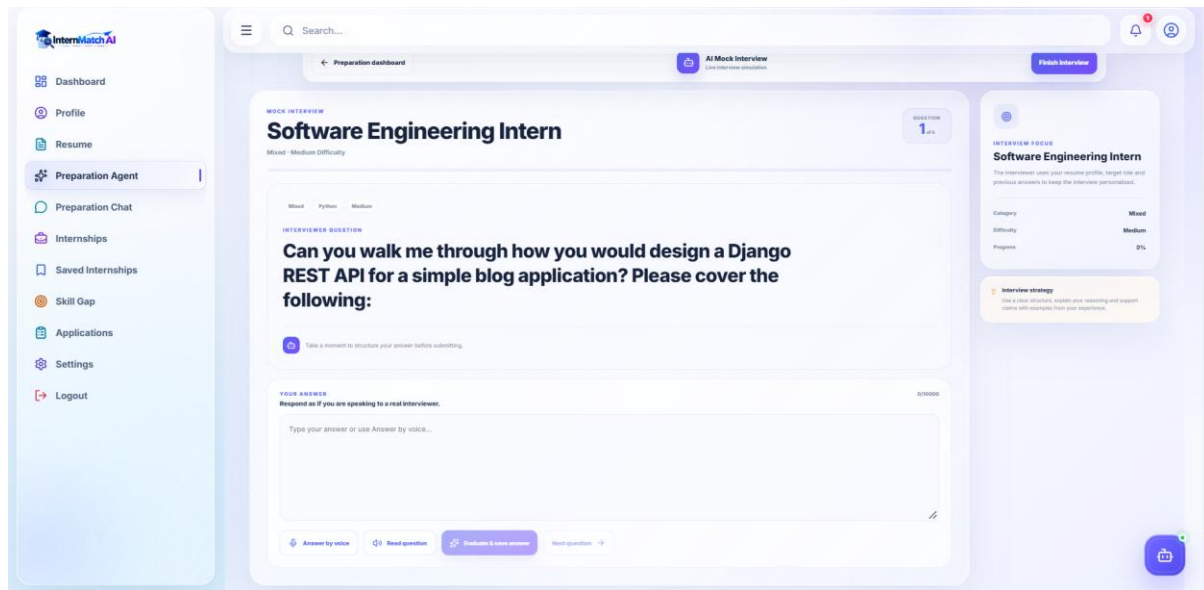
9.9 Preparation Agent

The Preparation Agent provides personalized interview preparation, generates interview questions, recommends preparation activities, and supports mock and voice interview workflows.

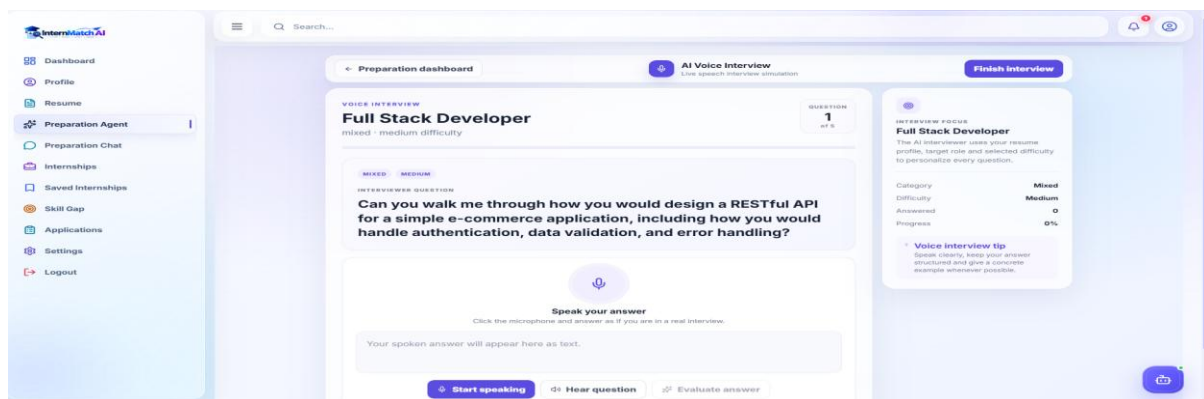




The system conducts a simulated interview by generating questions and collecting the user's answers. The session is used for interview practice and evaluation.

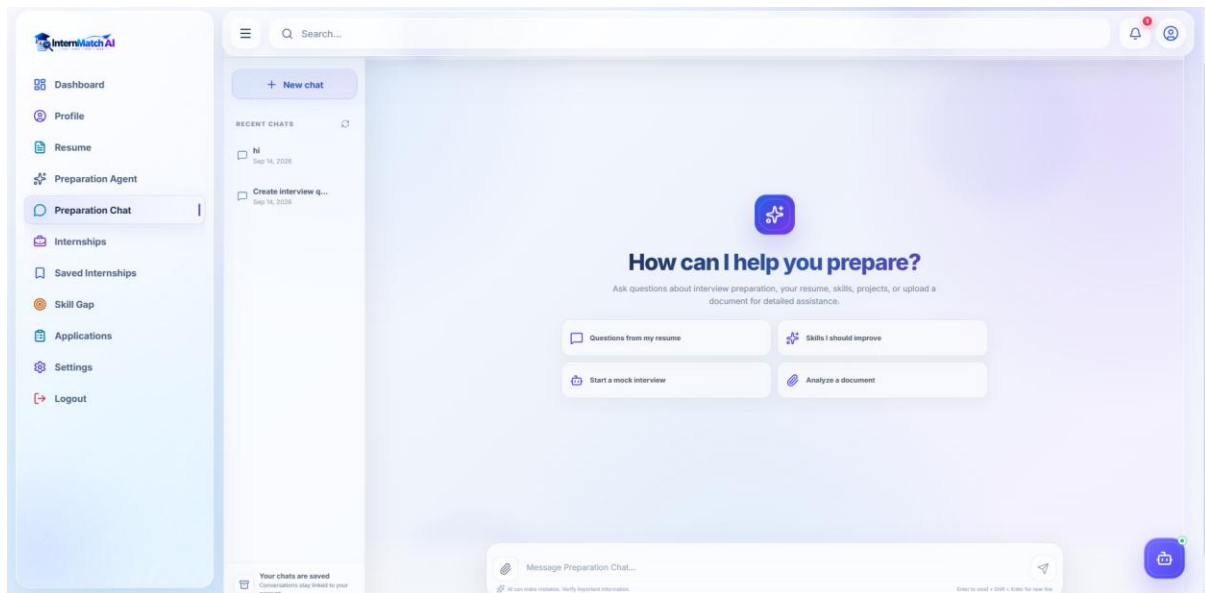


Users can participate in a voice-based interview, submit their answers, and receive AI-based evaluation of their responses.



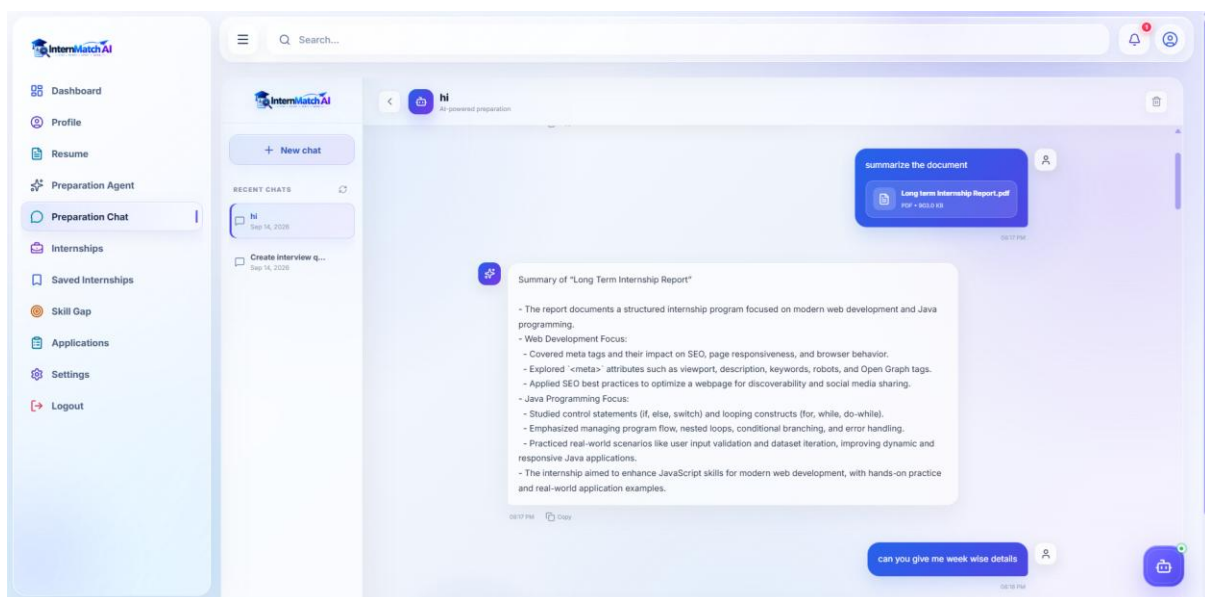
9.10 Preparation Chat

The Preparation Chat provides a ChatGPT-like interface for interview and career preparation. Users can create conversations, view conversation history, and delete conversations.



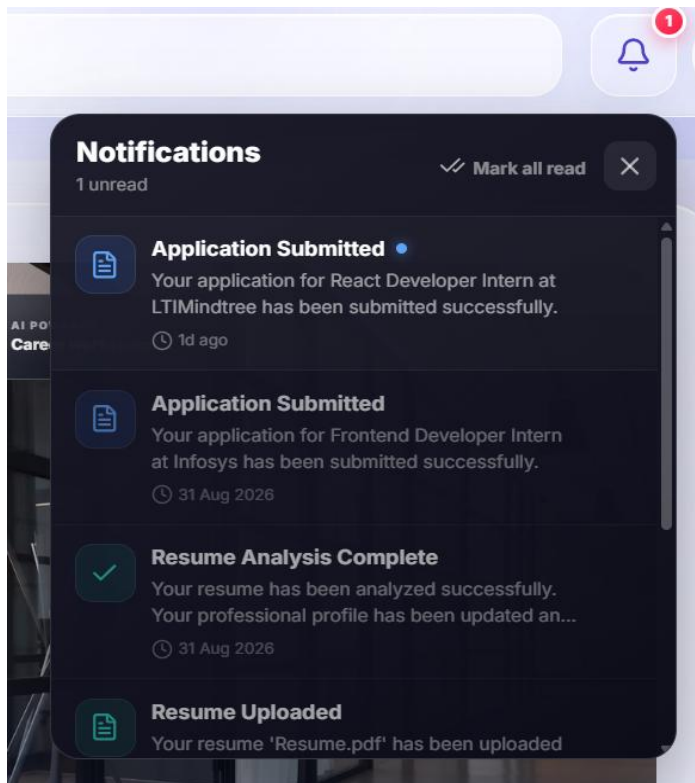
9.11 Preparation document upload and Analysis

Users can upload preparation documents in supported formats. The system extracts and processes the document into sections and chunks and uses the document content to provide contextual preparation responses.



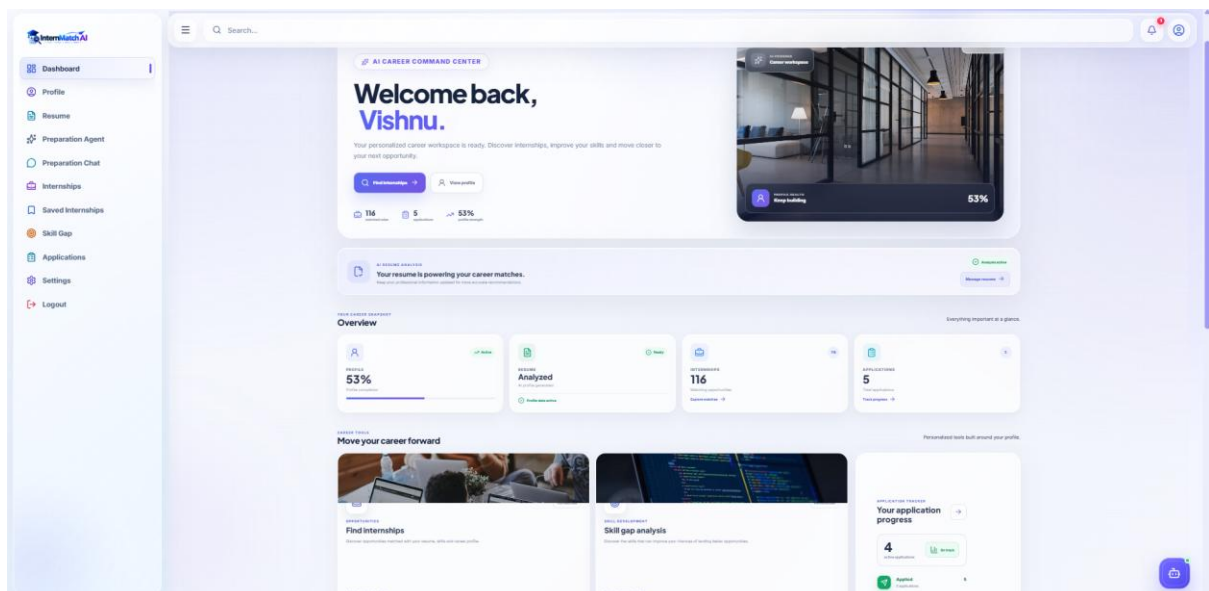
9.12 Notifications

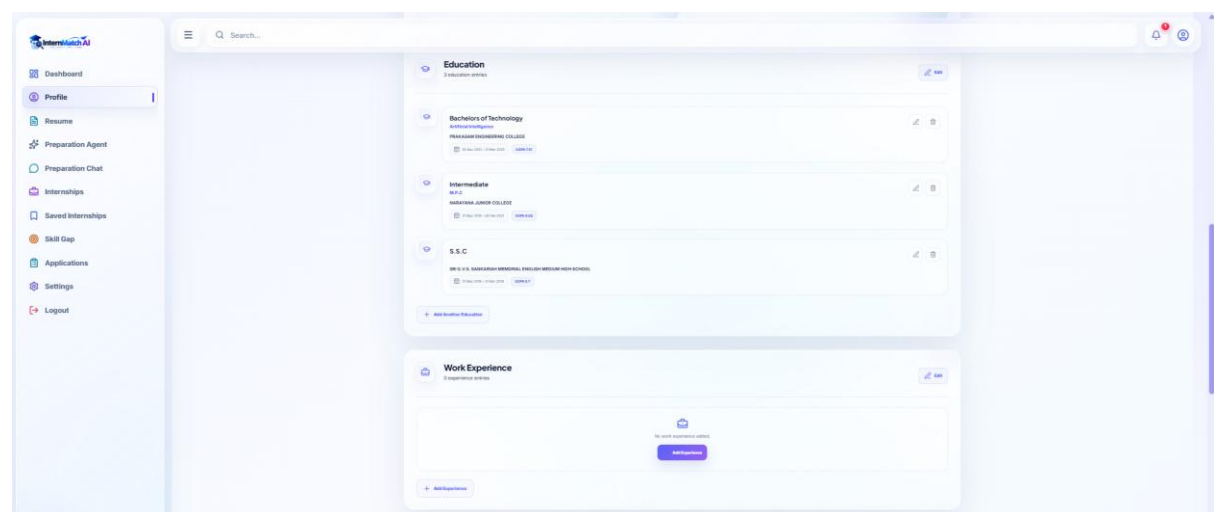
The system provides notifications related to relevant user activities and application events.

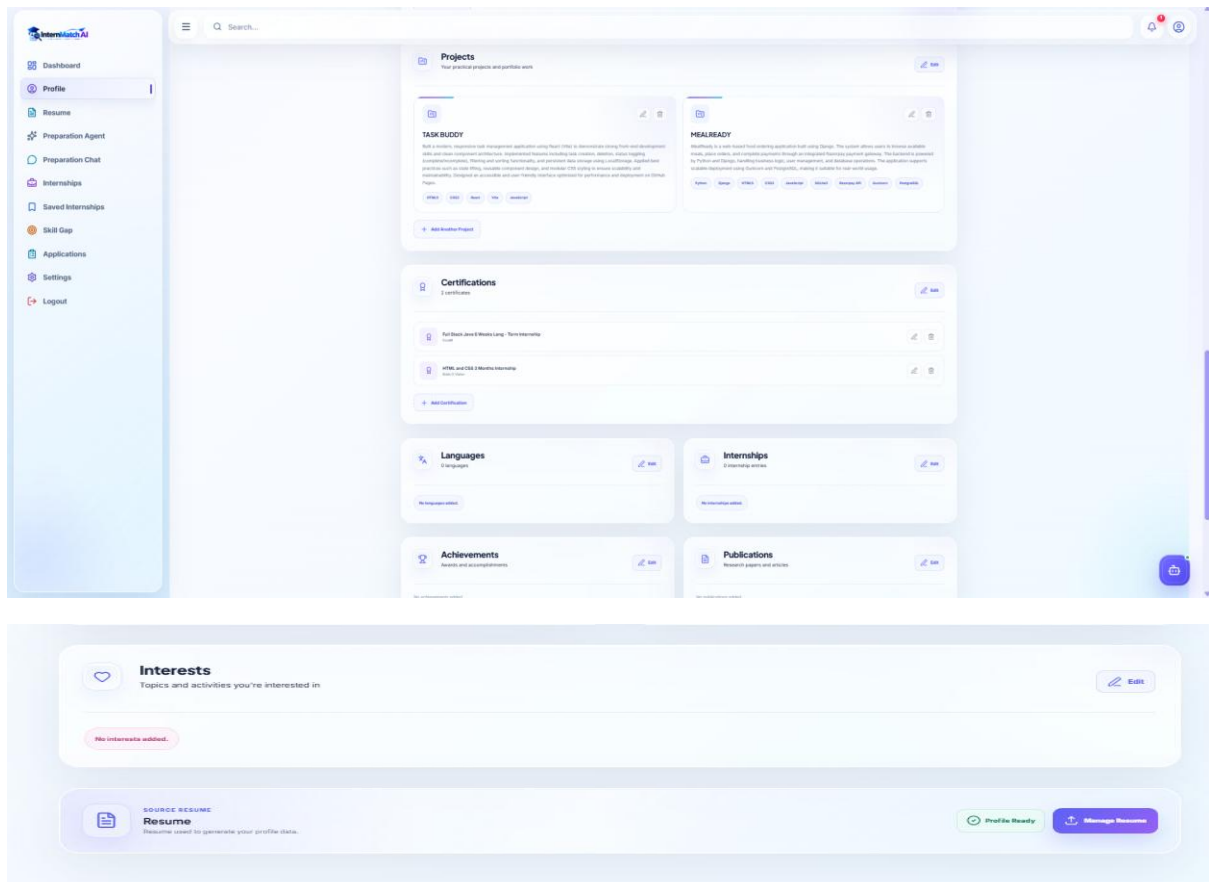


9.13 User dashboard and Profile

The dashboard provides access to the major internship, resume, application, AI, and preparation features. The profile displays the user's structured career information.

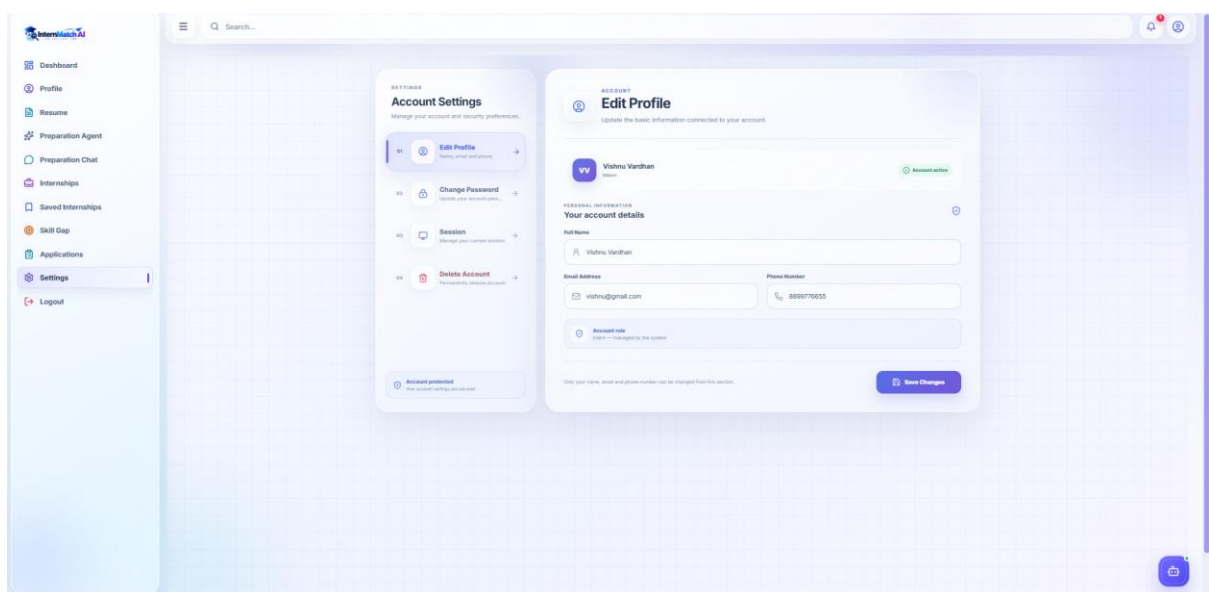






9.14 Settings Page

The Settings page allows users to manage their account and application preferences from a centralized interface. Users can review and update available account settings and control their application experience.



10. Testing

Testing was performed to verify the functional behavior of the **InternMatch AI** application across the frontend, backend APIs, database operations, AI-assisted features, internship workflows, and interview-preparation modules. The test cases below cover the major workflows implemented in the application.

Test ID	Test Case	Expected Result	Actual / Implementation Result	Status
T001	Register a new user with valid details	User account is created successfully	Registration workflow creates the user account and stores user data in PostgreSQL	Pass
T002	Login with valid credentials	User is authenticated and redirected to the appropriate dashboard	JWT authentication is performed and protected routes become accessible	Pass
T003	Upload a valid PDF/DOCX resume	Resume is accepted and processed	Resume file is validated and text extraction is performed	Pass
T004	Analyze uploaded resume	Resume information is converted into structured profile data	Regex parsing and Groq-based resume analysis generate structured resume information	Pass
T005	Search and filter internships	Relevant internships are displayed according to selected criteria	Internship listing, search, and filtering functionality is implemented	Pass
T006	View internship match and skill gap	User can view matching information and required skill gaps	Resume-based internship matching and skill-gap information are displayed	Pass
T007	Save and apply for an internship	Internship is saved or application is submitted successfully	Saved internship and application workflows persist data in PostgreSQL	Pass
T008	Generate an AI cover letter	Personalized cover letter is generated using available user and internship information	Groq-based cover-letter generation produces and stores the generated content	Pass

Test ID	Test Case	Expected Result	Actual / Implementation Result	Status
T009	Use AI Assistant	AI Assistant provides a contextual career-related response and maintains conversation history	AI Assistant processes available user/platform context and stores chat history	Pass
T010	Use Preparation Agent for interview preparation	User receives preparation guidance and generated interview questions	Preparation Agent provides interview preparation and question-generation functionality	Pass
T011	Start a mock interview	Interview session starts and questions are presented to the user	Mock interview workflow creates and manages interview session data	Pass
T012	Complete a voice interview and submit an answer	Spoken response is processed and evaluated	Voice interview workflow supports answer submission and AI-based answer evaluation	Pass
T013	Start a Preparation Chat conversation	User can send preparation questions and receive AI responses	Preparation conversations and messages are stored and retrieved successfully	Pass
T014	Upload a preparation document and ask questions	Uploaded document is processed and relevant information can be used in the response	PDF/DOCX document processing, sections, chunks, and document-message relationships are implemented	Pass
T015	View and delete Preparation Chat history	Stored conversations are displayed and selected conversations can be deleted	Conversation history retrieval and deletion functionality are implemented	Pass
T016	Access protected application features without authentication	Unauthorized user should not access protected resources	JWT-protected routes and backend authentication dependencies restrict unauthorized access	Pass

Testing Types

- **Functional Testing** — Registration, login, resume processing, internship search, matching, applications, cover letters, AI Assistant, Preparation Agent, interviews, and Preparation Chat workflows.
- **API Testing** — FastAPI endpoints, request validation, authentication, resume processing, internship, application, AI, and preparation APIs.
- **UI Testing** — React page navigation, forms, buttons, sidebars, dashboards, chat interfaces, interview interfaces, and responsive layouts.
- **Integration Testing** — Verification of communication between React frontend, FastAPI backend, PostgreSQL database, and Groq AI services.
- **Database Testing** — Verification of user, resume, internship, application, saved internship, chat, preparation, document, and interview data persistence and relationships.
- **AI Feature Testing** — Verification of resume analysis, cover-letter generation, AI Assistant responses, preparation responses, interview question generation, and answer evaluation.
- **File Processing Testing** — Verification of PDF/DOCX resume and preparation-document upload, text extraction, and document processing.
- **Authentication Testing** — Verification of JWT authentication, protected routes, password recovery, and user-specific access.

11. CHALLENGES AND SOLUTIONS

During the development of the **InternMatch AI** application, several technical challenges were encountered while integrating resume processing, internship management, AI services, interview preparation, and document-based preparation features. The major challenges and implemented solutions are summarized below.

Challenge	Solution
Different resume formats and structures	Implemented PDF/DOCX text extraction using PyMuPDF and python-docx, followed by Regex parsing and Groq LLM analysis.
Unstructured resume information	Combined Regex extraction with Groq-based structured analysis to create a consistent resume profile.
Secure authentication and protected pages	Implemented JWT authentication, password hashing, protected backend routes, and authentication dependencies.
Managing user-specific internships and applications	Used PostgreSQL and SQLAlchemy relationships to maintain users, internships, saved internships, and applications.
Preventing duplicate internship applications	Applied a unique user–internship constraint to prevent duplicate applications.
Generating personalized AI responses	Provided relevant resume, internship, and user context to Groq before generating AI responses.
Maintaining AI Assistant conversations	Implemented persistent chat sessions and conversation-history storage for authenticated users.
Building multiple interview-preparation features	Developed the Preparation Agent as a separate module containing preparation guidance, question generation, mock interviews, voice interviews, and AI evaluation.
Processing uploaded preparation documents	Implemented document extraction and processing with preparation documents, sections, chunks, and message-document relationships.
Integrating multiple frontend and backend modules	Connected React components with FastAPI REST APIs using Axios and organized functionality into separate backend routers and services.
Maintaining a consistent user experience	Implemented reusable React layouts, sidebar components, UI components, and responsive CSS.

12. FINAL PROJECT STATUS

12.1 Completed Features

- User registration, login, logout, JWT authentication, and protected routes.
- Password recovery and account management.
- PDF/DOCX resume upload, validation, and text extraction.
- Hybrid Regex and Groq LLM-based resume analysis.
- Structured resume profile with education, skills, experience, projects, certifications, and achievements.
- Internship catalog with search and filtering.
- Resume-based internship matching and match details.
- Skill gap analysis based on internship requirements.
- Saved internship management.
- Internship application and withdrawal functionality.
- Duplicate application prevention.
- AI-generated personalized cover letters with PDF/DOCX download.
- User notification functionality.
- AI Assistant with contextual responses and persistent chat history.
- Separate Preparation Agent with personalized interview preparation.
- Interview question generation and preparation recommendations.
- Mock Interview functionality.
- Voice Interview with AI-based answer evaluation.
- Preparation Chat with conversation history and conversation management.
- Preparation document upload and document-based AI responses.
- Preparation document sections, chunks, and message-document relationships.
- Responsive React interface with reusable layouts, sidebar components, and application navigation.

12.2 Final Submission Status

- Project implementation completed.
- Frontend and backend integration completed.
- PostgreSQL database and SQLAlchemy persistence implemented.
- AI features integrated using Groq LLM.
- Final functional testing completed.
- Final documentation prepared.
- Project demonstration prepared.

12.3 Final Outcome

The project implements an integrated **AI-powered internship matching and interview preparation platform** that supports users from resume analysis and

internship discovery through application support and interview preparation. The application combines resume intelligence, internship matching, skill-gap analysis, AI career assistance, personalized cover-letter generation, interview preparation, mock interviews, voice interviews, and document-based Preparation Chat in a single platform.

The application is implemented using a **React/Vite frontend**, **FastAPI backend**, **PostgreSQL database with SQLAlchemy**, and **Groq LLM services**. Resume and preparation documents are processed using appropriate PDF/DOCX extraction utilities, while structured information is maintained through the application's database models and relationships.

The **AI Assistant** provides contextual career and internship assistance, while the separate **Preparation Agent** provides interview-focused preparation, including mock and voice interview workflows. The **Preparation Chat** additionally allows users to upload preparation documents and receive context-based responses.

12.4 Conclusion

The completed project demonstrates the integration of **full-stack web development**, **secure authentication**, **resume processing**, **database management**, **internship matching**, **LLM-based generation**, **AI career assistance**, **interview preparation**, **voice interaction**, and **document processing** into a single internship-oriented application.

Project Status: Completed

Documentation Status: Completed

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Mentor: Vijay Parmar